10.4 - Integral Test for Convergence

Learning Objective(s):
LIM-7.A: Determine whether a series converges or diverges.

The Integral Test

THEOREM 10.4: THE INTEGRAL TEST

If f is positive, continuous, and decreasing for $x \geq m$ where m is a positive integer greater than or equal to 1 and $a_n = f(x)$, then

$$\sum_{n=1}^{\infty} a_n \text{ and } \int_1^{\infty} f(x)\,dx$$

either both converge or both diverge.

Example 1: Using the Integral Test
Confirm whether the Integral Test can be applied to the series. Then use the Integral Test to determine the convergence or divergence of the series.

a.) $\sum_{n=1}^{\infty} \frac{n}{n^2+1}$

diverges by the Integral Test

Consider $f(x) = \frac{x}{x^2+1}$.

$f(x) > 0$

$f(x)$ is continuous

$f'(x) = \frac{(x^2+1)\cdot 1 - x\cdot 2x}{(x^2+1)^2} = \frac{1-x^2}{(x^2+1)^2} < 0$

$\int_1^{\infty} \frac{x}{x^2+1}\,dx = \lim_{b\to\infty} \int_1^{b} \frac{x}{x^2+1}\,dx$

$u = x^2+1$
$du = 2x\,dx$

$= \frac{1}{2}\lim_{b\to\infty} \int_2^{b^2+1} \frac{du}{u}$

$= \frac{1}{2}\lim_{b\to\infty} \left[\ln|u|\right]_2^{b^2+1}$

$= \frac{1}{2}\lim_{b\to\infty} \left(\ln|b^2+1| - \ln 2\right) = \infty$

Diverges.

1. The Integral Test will always result in having to evaluate an improper integral (Topic 6.13).
2. The AP Calculus BC exam has required students to state the conditions for which the Integral Test can be applied. You typically will not have to *prove* they apply, rather, just state that they apply.

SCAN ME

Scan the QR Code above to watch a video covering Example 1a

<u>Example 1 (cont'd)</u>: Using the Integral Test

Confirm whether the Integral Test can be applied to the series. Then use the Integral Test to determine the convergence or divergence of the series.

b.) $\sum_{n=3}^{\infty} \frac{n}{\sqrt{2n-5}}$

Consider $f(x) = \frac{x}{\sqrt{2x-5}}$

Positive ✓

Continuous ✓

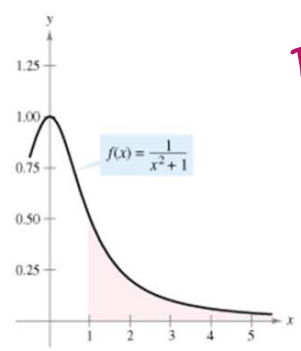
Decreasing X

$\lim_{n\to\infty} \frac{n}{\sqrt{2n-5}} = \infty$

$\sum_{n=3}^{\infty} \frac{n}{\sqrt{2n-5}}$ diverges by the nth term test!

Scan the QR Code above to watch a video covering Example 1b

c.) $\sum_{n=1}^{\infty} \frac{1}{n^2+1}$ converges by the Integral Test!



P ✓

C ✓

D ✓

Consider $\int_1^{\infty} \frac{1}{x^2+1}\,dx$

$= \lim_{b\to\infty} \int_1^b \frac{1}{x^2+1}\,dx$

$= \lim_{b\to\infty} \left[\tan^{-1} x\right]_1^b$

$= \lim_{b\to\infty} \left(\tan^{-1} b - \tan^{-1} 1\right)$

$= \frac{\pi}{2} - \frac{\pi}{4} = \frac{\pi}{4}$

Scan the QR Code above to watch a video covering Example 1c

10.5 - Harmonic Series and p-Series

Learning Objective(s):
LIM-7.A: Determine whether a series converges or diverges.

This Topic will introduce one of the most common series, the ***p*-series**. Fortunately, it is perhaps the simplest type of series to determine convergence or divergence.

MATHEMATICS & HISTORY

PYTHAGORAS (c. 570 – c. 495 BC)
In addition to his countless contributions to geometry, Pythagoras and his students paid close attention to the development of music as an abstract science. This led to the discovery of the relationship between the tone and the length of a vibrating string. It was observed that the most beautiful musical harmonies corresponded to the simplest ratios of whole numbers. Later mathematicians developed this data into the harmonic series, where the terms in the series correspond to the nodes on a vibrating string that produce multiples of the fundamental frequency. For example, ½ is twice the fundamental frequency, 1/3 is three times the fundamental frequency, and so on.

The *p*-Series

DEFINITION *p*-SERIES

A *p*-series is an infinite series of the form

$$\sum_{n=1}^{\infty} \frac{1}{n^p} = \frac{1}{1^p} + \frac{1}{2^p} + \frac{1}{3^p} + \cdots + \frac{1}{n^p} + \ldots$$

where p is a positive number.

Example 1: Recognizing a *p*-Series.

Which of the following series are *p*-series? State yes or no.

✓ **a.)** $\sum_{k=1}^{\infty} k^{-0.8} = \sum_{k=1}^{\infty} \frac{1}{k^{0.8}}$

✗ **b.)** $\sum_{n=1}^{\infty} 2^{-n} = \sum_{n=1}^{\infty} \frac{1}{2^n}$

✓ **c.)** $\sum_{k=10}^{\infty} k^{-4} = \sum_{k=10}^{\infty} \frac{1}{k^4}$

✗ **d.)** $\sum_{n=1}^{\infty} n^2 = \sum_{n=1}^{\infty} \frac{1}{n^{-2}}$

✓ **e.)** $\sum_{k=1}^{\infty} \frac{1}{k^1}$

THEOREM 10.5-A: CONVERGENCE OF A *p*-SERIES

The *p*-series

$$\sum_{n=1}^{\infty} \frac{1}{n^p} = \frac{1}{1^p} + \frac{1}{2^p} + \frac{1}{3^p} + \cdots + \frac{1}{n^p} + \ldots$$

1. converges if $p > 1$, and
2. diverges if $0 < p \leq 1$.

SCAN ME

Scan the QR Code above to watch a video covering Example 1

Example 2: **Convergent and Divergent *p*-Series.**

Scan the QR Code above to watch a video covering Example 2

Determine whether the following series converge or diverge.

a.) $\sum_{k=1}^{\infty} \frac{1}{\sqrt[4]{k^3}} = \sum_{k=1}^{\infty} \frac{1}{k^{3/4}}$

$p = \frac{3}{4} \le 1$ Diverges!

b.) $\sum_{n=4}^{\infty} \frac{1}{(n-1)^2} = \frac{1}{9} + \frac{1}{16} + \frac{1}{25} + \cdots$

$p = 2 > 1$

Converges!

c.) $\sum_{k=2}^{\infty} \frac{1}{e^k}$

$r = \frac{1}{e} < 1$

$\frac{\frac{1}{e^2}}{1 - \frac{1}{e}} \cdot \frac{e^2}{e^2} = \frac{1}{e^2 - e}$

Converges!

Example 3: **Convergent and Divergent *p*-Series.**

Determine the convergence or divergence of the following *p*-series.

$$1 + \frac{1}{\sqrt[3]{4}} + \frac{1}{\sqrt[3]{9}} + \frac{1}{\sqrt[3]{16}} + \frac{1}{\sqrt[3]{25}} + \cdots = \sum_{k=1}^{\infty} \frac{1}{k^{2/3}}$$

$2^2 \quad 3^2 \quad 4^2 \quad 5^2$

$\frac{2}{3} \le 1$

Divergent p-series!

The Harmonic Series

Scan the QR Code above to watch a video covering Example 3 & The Harmonic Series

DEFINITION HARMONIC SERIES

A harmonic series is an infinite series of the form

$$\sum_{n=1}^{\infty} \frac{1}{n} = \frac{1}{1} + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{n} + \ldots$$

Note: A harmonic series is simply a *p*-series where the value of *p* is 1.

THEOREM 10.5-B: DIVERGENCE OF THE HARMONIC SERIES

The harmonic series

$$\sum_{n=1}^{\infty} \frac{1}{n} = \frac{1}{1} + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{n} + \ldots$$

diverges!

10.6 - Comparison Tests for Convergence

Learning Objective(s):
LIM-7.A: Determine whether a series converges or diverges.

Direct Comparison Test

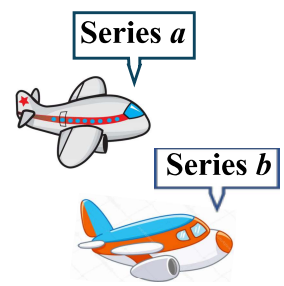
For the convergence tests that we have discussed so far, the terms of the series seem to be pretty simple and the series must have a special "look" to it in order for the convergence tests to be applied.
Any slight deviation from that "look" or characteristic, can make the convergence tests non-applicable.

1. $\sum_{n=0}^{\infty} \frac{1}{2^n}$ is geometric, but $\sum_{n=0}^{\infty} \frac{n}{2^n}$ is not.

2. $\sum_{n=1}^{\infty} \frac{1}{n^3}$ is a p-series, but $\sum_{n=0}^{\infty} \frac{1}{n^3+1}$ is not.

3. For $\sum_{n=1}^{\infty} \frac{n}{\left(n^2+3\right)^2}$, $a_n = \frac{n}{\left(n^2+3\right)^2}$ is easily integrated, but For $\sum_{n=1}^{\infty} \frac{n^3}{\left(n^2+3\right)^2}$, $b_n = \frac{n^3}{\left(n^2+3\right)^2}$ is not.

Our goal now becomes to investigate a way to **compare** a series with complicated terms with a simpler series whose convergence or divergence is known.



THEOREM 10.6-A: THE DIRECT COMPARISON TEST

Let $0 < a_n \le b_n$ for all n.

1. If $\sum_{n=1}^{\infty} b_n$ converges, then $\sum_{n=1}^{\infty} a_n$ converges.

2. If $\sum_{n=1}^{\infty} a_n$ diverges, then $\sum_{n=1}^{\infty} b_n$ diverges.

SCAN ME

Scan the QR Code above to watch a video covering Example 1

Example 1: Using the Direct Comparison Test
Determine the convergence or divergence of the following series.

a.) $\sum_{n=1}^{\infty} \frac{1}{2+3^n} = \sum_{n=1}^{\infty} a_n$, where $a_n = \frac{1}{2+3^n}$.

converges by the DCT!

Let $b_n = \frac{1}{3^n}$.

$\sum_{n=1}^{\infty} \frac{1}{3^n}$ converges!

$\left|\frac{1}{3}\right| < 1$

$2+3^n > 3^n$

$\frac{1}{2+3^n} < \frac{1}{3^n}$

$a_n < b_n$

b.) $\sum_{n=1}^{\infty} \frac{1}{2+\sqrt{n}}$ $\underbrace{}_{a_n}$

diverges by DCT!

$2+\sqrt{n} < n$

$\frac{1}{2+\sqrt{n}} > \frac{1}{n}$ for $n > 4$

$b_n = \frac{1}{n}$

$a_n > b_n$

$\sum_{n=1}^{\infty} b_n$ diverges

Remember that both parts of the Direct Comparison Test require that $0 < a_n \le b_n$.
Informally, the test says the following about the two series with nonnegative terms

1. **If the "larger" series converges, the "smaller" series must also converge.**
2. **If the "smaller" series diverges, then the "larger" series must also diverge.**

Limit Comparison Test

Often a series closely resembles a *p*-series or a geometric series, yet one cannot establish the term-by-term comparison necessary to apply the Direct Comparison Test.
If this happens, it may be possible to apply a second type of test called the **Limit Comparison Test.**

THEOREM 10.6-B: LIMIT COMPARISON TEST

Suppose $a_n > 0,\ b_n > 0$, and

$$\lim_{n\to\infty}\left(\frac{a_n}{b_n}\right) = L \quad \text{where } L \text{ is } \textit{finite and positive.}$$

Then the two series $\sum a_n$ and $\sum b_n$ either both converge, or both diverge.

Example 2: Using the Limit Comparison Test
Show that the following general harmonic series diverges.

$$\underbrace{\sum_{n=1}^{\infty}\frac{1}{an+b},\ a>0,\ b>0}_{\text{diverges by LCT}} \qquad \sum_{n=1}^{\infty}\frac{1}{n} \text{ diverges}$$

$$a_n = \frac{1}{an+b},\quad b_n = \frac{1}{n}$$

$$\lim_{n\to\infty}\frac{a_n}{b_n} = \lim_{n\to\infty}\frac{n}{an+b} = \frac{1}{a}$$

Helpful examples in choosing an appropriate *p*-series when using the Limit Comparison Test

General Series	Comparison Series	Conclusion
$\sum_{n=1}^{\infty}\frac{1}{3n^2-4n+5}$	$\sum_{n=1}^{\infty}\frac{1}{n^2}$	Both series converge.
$\sum_{n=1}^{\infty}\frac{1}{\sqrt{3n-2}}$	$\sum_{n=1}^{\infty}\frac{1}{\sqrt{n}}$	Both series diverge.
$\sum_{n=1}^{\infty}\frac{n^2-10}{4n^5+n^3}$	$\sum_{n=1}^{\infty}\frac{n^2}{n^5}=\sum_{n=1}^{\infty}\frac{1}{n^3}$	Both series converge.

Example 3: Using the Limit Comparison Test

Determine the convergence or divergence of the following series.

a.) $\sum_{n=1}^{\infty} \frac{\sqrt{n}}{n^2+1} = \sum_{n=1}^{\infty} a_n$, where $a_n = \frac{\sqrt{n}}{n^2+1}$.

Let $b_n = \frac{1}{n^{3/2}}$.

$\lim_{n\to\infty} \frac{a_n}{b_n} = \lim_{n\to\infty} \frac{n^{3/2} \cdot n^{1/2}}{n^2+1} = 1$

$\sum_{n=1}^{\infty} b_n$ converges (p-series w/ $p = \frac{3}{2} > 1$)

Converges by the LCT!

b.) $\sum_{n=1}^{\infty} \frac{n2^n}{4n^3+1} = \sum_{n=1}^{\infty} a_n$

diverges by LCT

$a_n = \frac{n \cdot 2^n}{4n^3+1}$ $\quad b_n = \frac{2^n}{n^2}$

$\lim_{n\to\infty} \frac{a_n}{b_n} = \lim_{n\to\infty} \frac{n \cdot 2^n \cdot n^2}{2^n(4n^3+1)} = 1/4$

$\sum_{n=1}^{\infty} \frac{2^n}{n^2}$ diverges

$\lim_{n\to\infty} \frac{2^n}{n^2} = \infty$

SCAN ME

Scan the QR Code above to watch a video covering Example 2 & 3

Polynomial Test

The following is a special case of the Comparison Tests from this section.
The Polynomial Test is not specifically tested on the AP Calculus BC exam nor will it be specifically referenced in this course, but it provides great insight in determining the behavior of certain series.

POLYNOMIAL TEST

Let $P(n)$ and $Q(n)$ be polynomials of degree j and k respectively.

1. $\sum_{n=1}^{\infty} \frac{P(n)}{Q(n)}$ converges if $j < k-1$.

2. $\sum_{n=1}^{\infty} \frac{P(n)}{Q(n)}$ diverges if $j \geq k-1$.

10.7 - Alternating Series Test for Convergence

Learning Objective(s):
LIM-7.A: Determine whether a series converges or diverges.

Alternating Series

Up to now, most series we have dealt with had positive terms. In this section and beyond, we'll study series that contain both positive and negative terms. The simplest of these series is called an **alternating series** whose terms alternate signs.

DEFINITION: ALTERNATING SERIES

An alternating series is a series either of the form

$$\sum_{n=1}^{\infty}(-1)^{n+1}a_n = a_1 - a_2 + a_3 - a_4 + \cdots \qquad \text{or}$$

$$\sum_{n=1}^{\infty}(-1)^{n}a_n = -a_1 + a_2 - a_3 + a_4 - \cdots$$

where $a_n > 0$ for all integers $n \geq 1$.

MATHEMATICS & HISTORY

GOTTFRIED WILHELM LEIBNIZ
(1646-1716)
Leibniz is a German mathematician and philosopher who is credited as one of the two main discoverers of calculus having worked indepent of Isaac Newton. In addition to developing the common notation for derivatives, dy/dx, that we still use today, Leibniz developed the Alternating Series test which is sometimes called the Leibniz Test or Leibniz Criterion.

THEOREM 10.7: ALTERNATING SERIES TEST

Let $a_n > 0$. The alternating series

$$\sum_{n=1}^{\infty}(-1)^{n}a_n \quad \text{and} \quad \sum_{n=1}^{\infty}(-1)^{n+1}a_n$$

converge if the following two conditions are met

1. $\lim_{n\to\infty} a_n = 0$ **2.** If $a_{n+1} \leq a_n$ for all n

Note: Another way to state Criterion #2 above is that the a_n values are **nonincreasing**.

SCAN ME

Scan the QR Code above to watch a video covering Example 1

Example 1: Using the Alternating Series Test

Determine the convergence or divergence of the following series.

a.) $\sum_{n=1}^{\infty}(-1)^{n+1}\frac{1}{n} = \sum_{n=1}^{\infty}(-1)^{n+1}a_n$ converges by the AST.

$a_n = \frac{1}{n}$

$\lim_{n\to\infty}\frac{1}{n} = 0$

$a_{n+1} - a_n = \frac{1}{n+1} - \frac{1}{n} = \frac{n-(n+1)}{n(n+1)} = \frac{-1}{n(n+1)} \leq 0$

b.) $\sum_{n=1}^{\infty}(-1)^{n}\frac{n}{n+1} = \sum_{n=1}^{\infty}(-1)^{n}a_n$ diverges by the nth term test

$a_n = \frac{n}{n+1}$

$\lim_{n\to\infty}\frac{n}{n+1} = 1 \neq 0$

VERIFYING NONINCREASING BEHAVIOR FOR A SEQUENCE

1. Use the **Algebraic Difference** test to show that $a_{n+1} - a_n \le 0$ for all $n \ge 1$.

2. Use the **Algebraic Ratio** test to show that $\frac{a_{n+1}}{a_n} \le 1$ for all $n \ge 1$.

3. Use the **derivative** of the related function, $f(x)$, for which $a_n = f(n)$ for all n, defined for $x > 0$ and show that $f'(x) \le 0$ for all $x \ge 0$.

Example 2: Showing an Alternating Series Converges

Show that the following alternating series converges.

$$\sum_{n=1}^{\infty} \frac{(-1)^n}{(2n)!} = \underbrace{\sum_{n=1}^{\infty} (-1)^n a_n}_{\text{converges by the AST!}}, \text{ where } a_n = \frac{1}{(2n)!}$$

$$\lim_{n\to\infty} \frac{1}{(2n)!} = 0$$

$$\frac{a_{n+1}}{a_n} = \frac{(2n)!}{(2n+2)!} = \frac{1}{(2n+2)(2n+1)} \le 1$$

$$(2n+2)! = (2n+2)(2n+1)(2n)!$$

SCAN ME

Scan the QR Code above to watch a video covering Example 2